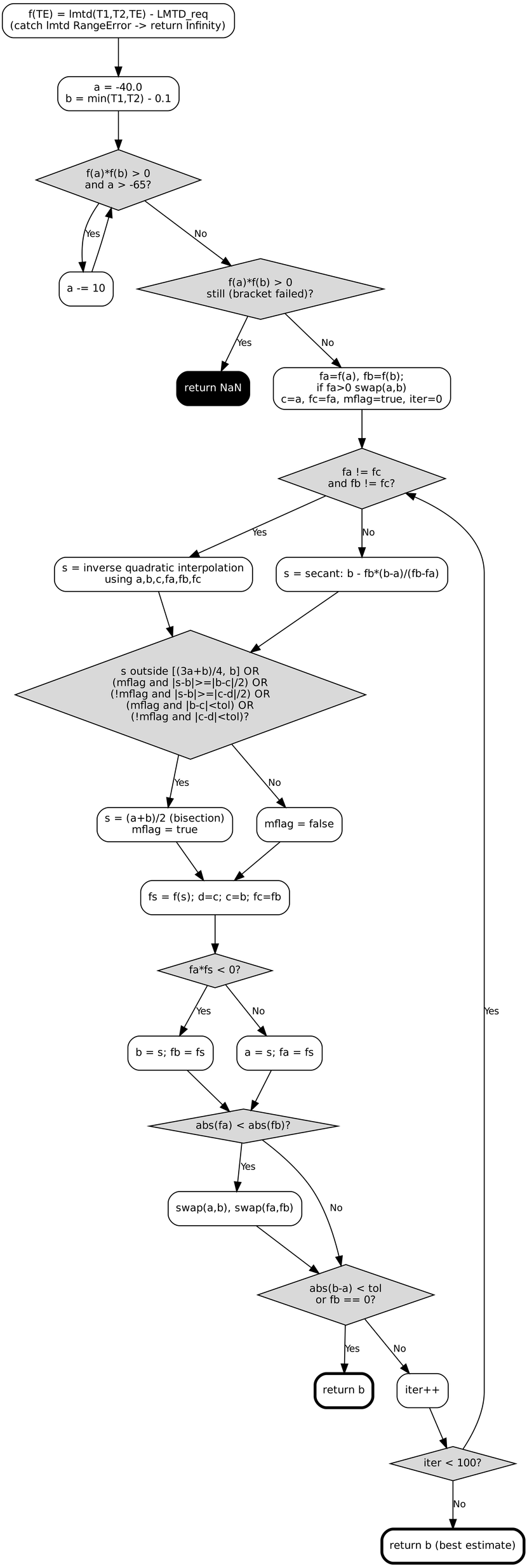
f(TE) = lmtd(T1,T2,TE) - LMTD_req
(catch lmtd RangeError -> return Infinity)
a = -40.0
b = min(T1,T2) - 0.1
f(a)*f(b) > 0
and a > -65?
Yes
No
a -= 10
f(a)*f(b) > 0
still (bracket failed)?
Yes
No
return NaN
fa=f(a), fb=f(b);
if fa>0 swap(a,b)
c=a, fc=fa, mflag=true, iter=0
fa != fc
and fb != fc?
Yes
No
s = inverse quadratic interpolation
using a,b,c,fa,fb,fc
s = secant: b - fb*(b-a)/(fb-fa)
s outside [(3a+b)/4, b] OR
(mflag and |s-b|>=|b-c|/2) OR
(!mflag and |s-b|>=|c-d|/2) OR
(mflag and |b-c|<tol) OR
(!mflag and |c-d|<tol)?
Yes
No
s = (a+b)/2 (bisection)
mflag = true
mflag = false
fs = f(s); d=c; c=b; fc=fb
fa*fs < 0?
Yes
No
b = s; fb = fs
a = s; fa = fs
abs(fa) < abs(fb)?
Yes
No
swap(a,b), swap(fa,fb)
abs(b-a) < tol
or fb == 0?
Yes
No
return b
iter++
iter < 100?
Yes
No
return b (best estimate)